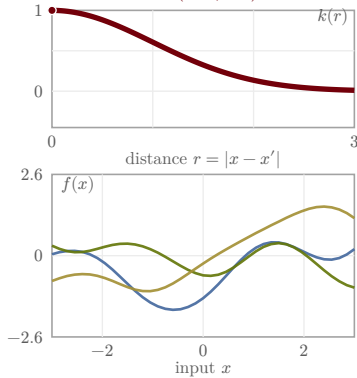


Gallery of GP covariance kernels

length-scale $\ell = 1$, signal variance $\sigma_f^2 = 1$ (period $p = 2$, RQ mixing $\alpha = 1$)

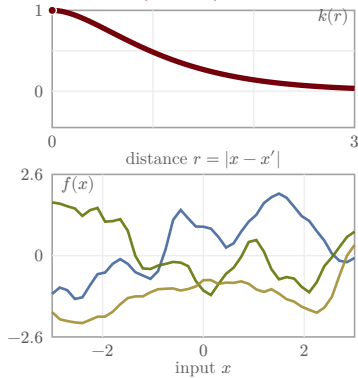
squared-exp.

$$\exp(-r^2/2\ell^2)$$



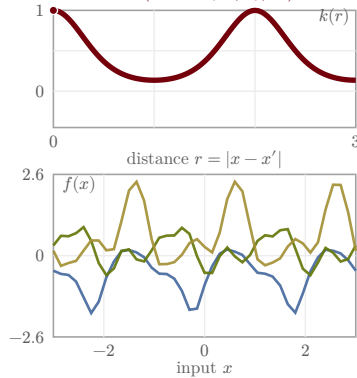
Matérn $\nu = 3/2$

$$(1 + \sqrt{3}r)e^{-\sqrt{3}r}$$



periodic

$$\exp(-2 \sin^2(\pi r/p)/\ell^2)$$



rational-quad.

$$(1 + r^2/(2\alpha))^{-\alpha}$$

